



PROJECT REPORT

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Submitted
jointly with the
project's
source files.

INTRODUCTION

#INCLUDE

Introduction

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Abstract: Faced
with the task
to create a
semi-functional
banking
management
system, we go
through all the
necessary steps
to make our
vision a reality
from pure
conception to
absolute
conviction.

ON A BEAUTIFUL SUNNY DAY AT THE HIGH MOUNTAIN WHERE THE EVEN HIGHER INSTITUTE OF DIGITAL ENGINEERING RESIDED, PROFESSOR BOUSSLAMA RIHAB GAVE HER STUDENTS A PROJECT TO REALIZE AS A MEANS OF TESTING NOT ONLY WHAT THEY HAVE COVERED IN CLASS PREVIOUSLY, BUT ALSO THEIR TEAM SPIRIT, COORDINATION AND RESILIENCE BEFORE LONG CHALLENGING TASKS.

THE GAME TASK WAS AS FOLLOWS: " THE PROJECT AIMS TO DEVELOP A BANKING MANAGEMENT SYSTEM CAPABLE OF MANAGING EMPLOYEE RECORDS, CUSTOMER ACCOUNTS, FINANCIAL TRANSACTIONS, BANK BRANCHES, AND LOAN OPERATIONS." ALONG WITH AN EXTENSIVE GUIDE FOR GUIDELINES AND A DEADLINE.

A MONTH LATER, THIS GROUP CELEBRATES THE COMPLETION OF THE PROJECT AND, AT THE TIME OF LECTURE, HAS ALREADY SUBMITTED THIS REPORT AFTER EXERCISING DUE DILIGENCE IN ACCORDANCE WITH THE GIVEN TASKS.

AS SUCH, THIS REPORT WILL COVER THE AUTHORS' METHODOLOGY IN DECONSTRUCTING THE GIVEN PROBLEM AND THE FINAL PRODUCT VISION. AN EXPLANATION OF NOTABLE CHOICES WILL BE MADE CLEAR AND PURPOSEFUL.

#DEFINE "METHODOLOGY"

FIRST CONTACT () {



STUDENTS IN GENERAL WERE LEFT IN AWE FROM THE NATURE OF THE PROJECT. IT SEEMED LONG, HARD, UNDECIPHERABLE EVEN. OUR GROUP WAS NONE THE WISER. AFTER ALL, WE WERE NOT FAMILIAR WITH COMPUTER SCIENCE DEMANDING TASKS. EVEN WORSE, SOME CONCEPTS DECLARED IN THE PROJECT COVERED TOPICS WE STILL DIDN'T STUDY AT THE TIME. IT IS SAFE TO SAY THAT WE WERE COMPLETELY OBLIVIOUS TO HOW TO DECONSTRUCT THAT BEHEMOTH OF A TASK. WE WERE UNABLE TO GO BEYOND THE CREATION OF A REPOSITORY ON "GITHUB".

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FIRST APPROACH () {

WITH TIME RUNNING AS FAST AS IT COULD, PERHAPS A SIDE EFFECT OF PEER PRESSURE, WE DECIDED TO HOLD OUR FIRST MEETING TO DISCUSS MATTERS AND DETERMINE OBJECTIVES. AND HOW FRUITFUL OF A MEETING THAT WAS. WE UNDERLINED WHAT WE COULD DO, AND WHAT WE STILL DON'T KNOW. NAMELY, WE KNEW THAT ORGANIZING STRUCTURES SHOULD BE OUR STARTING POINT. WITH THAT IN MIND, WE DIVIDED THE GROUP WORK AND DISCUSSED OUR INITIAL PLAN, TO READY THE LINEAR DATA STRUCTURES REQUIRED IN THE PROJECT LEAVING, AT THE TIME, "STACKS AND QUEUES" FOR LATER UPDATES. IN ADDITION, WE AGREED ON VARIABLES' TYPES (SOME CHANGES WERE MADE IN LATER PLANS) BUT ALL IN ALL, WE HAD A GOOD HEADSTART. AT THIS POINT, WE STILL DIDN'T HAVE A CLEAR VIEW OF HOW WE SHOULD MAKE THE REQUIRED INTERFACE ALTHOUGH ONE THING WAS CERTAIN, WE HAD TO QUICKLY DETERMINE OUR STANDARDS FOR INPUTS.

FIRST VISION () {

RAW FEELINGS FROM PURE PEER PRESSURE WERE ALWAYS AT FULL EFFECT. NOT LONG AFTER DISCUSSING OUR MAIN IDEAS ABOUT THE PROJECT, I REALISED THAT WE WERE STILL MISSING THE LINK THAT WILL ENABLE US TO CONNECT EVERYTHING TOGETHER. THERE WAS ALSO THE CONCERN THAT GROUP WORK SHOULD BE DONE IN COORDINATION, AND THAT ANY MISCOMMUNICATION COULD EASILY HINDER THE WORK OF THE GROUP. IN ORDER TO ADVANCE THE PROJECT FURTHER, AND FOR THE SAKE OF THE GROUP, THAT DAY, I CREATED THE "EXTENSIVE PROJECT PLAN" (REFERRED TO AS EPP FROM NOW ON), HIGHLIGHTING NECESSARY ACTIONS BY DECONSTRUCTING THE PROJECT DOWN TO ITS MINUTE DETAILS. IT DIDN'T MATTER THAT WE STILL DIDN'T COVER EVERYTHING IN CLASS BECAUSE THE SCHEME MADE US IN A READY STATE FOR STRIKING AT ANY MOMENT.

AT THE SAME TIME, MY TEAMMATES SHOWED SOME CONCERNS ABOUT THE SLOW ADVANCE OF THE PROJECT. FORTUNATELY, GIVEN THE PLAN, IT SEEMED THAT ALL OF US KNEW EXACTLY WHAT WE WERE SUPPOSED TO DO. THAT MARKED OUR FINAL ROLE DIVISION FOR THE GIVEN TASKS. IT IS NOW HIGH TIME TO GO ON THE OFFENSIVE.

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THE EPP WASN'T PERFECT IN ANY WAY, BUT IT WAS A SIGNIFICANT ADVANCEMENT THAT WE LEVERAGED TO DO MOST OF THE PROJECT. HERE ARE THE MAIN POINTS IT HIGHLIGHTED:

- FIRSTLY, IT WAS PRESENTED AS AN EXTENSIVELY ANNOTATED MIND MAP, FOLLOWING A CHOICE SHEET MINDSET DESCRIBING HOW AN ACTUAL USER WOULD INTERACT WITH TASK ELEMENTS AND TASK ORDERS. THAT WAY, THE EPP MADE INTERACTIONS BETWEEN PROJECT ELEMENTS CLEAR USING VISUALS. WE WERE IN FACT AWARE THAT ALMOST ANY INTERACTION WOULD RESULT IN A VISUAL CHANGE, AND THE SAVE OR DISPLAY OF A CERTAIN VALUE.

- GREAT CARE WAS ALLOCATED IN DECIDING THE HIERARCHY OF THE ELEMENTS IN THE TASKS.

- REQUIRED STRUCTURES WERE CITED FOR TRACKING PURPOSES.

- INPUTS WERE COLOR CODED FOR EASIER IDENTIFICATION. IN ADDITION, DIFFERENT TYPES HAD THEIR OWN DISTINCTIVE COLORS.

- WE INTRODUCED THE FOLLOWING NOTATIONS:

- "@UTO" FOR VARIABLES THAT REQUIRE NO REALTIME USER INPUT. THESE VARIABLES ARE GENERATED BY THE MACHINE FOLLOWING A SET OF STANDARDS.

- "***" FOR DENOTING UNIQUE VARIABLES REQUIRING A "UNIQUENESS CHECK" IN THEIR RESPECTIVE SET OF ELEMENTS.

- "@TRIG" TO INDICATE THAT FOR EACH CHOICE, A CORRESPONDING ACTION WOULD OCCUR. AS IMPERFECT AS IT LOOKED ON PAPER, IT ACTUALLY SERVED AT MAKING THE GROUP MORE AWARE OF CALLING FUNCTIONS WHEN IMPLEMENTING ACTUAL CODE.

- ONE OF THE MOST IMPORTANT ASPECTS OF THIS SHEET IS THE FACT THAT IT STANDARDIZED INPUTS SO THAT WE ARE ALL ON THE SAME PAGE. NAMELY:

- TIME CYCLES: WE REALISED THAT WHILE INCLUDING A "TIMEKEEPING" FUNCTION WOULD WORK, NOT ONLY WOULD IT BE HIGHLY INEFFICIENT FOR TESTING PURPOSES, BUT ALSO IT WOULD BREAK THE SPIRIT OF THE RULES THAT HINTED TOWARDS MAKING THE DAY/NIGHT CYCLE RELATED TO THE LOG IN/OUT CYCLE OF THE EMPLOYEE.

- ACCOUNT'S ATTRIBUTE REGULATION, MAINLY, WE PUT A STANDARD ON CUSTOMERS' ACCOUNT NUMBERS, THEY SHALL BE MADE USING A 12-CHARACTER STRING FROM A 54-BASE MIX OF LETTERS AND NUMBERS. OUR REASONING IS THAT, AS IT TURNED OUT, BANKS MANAGE THE EQUIVALENT OF MILLIONS OF ACCOUNTS. TO ENSURE CONSISTENCY AND TO GENERATE A MODERATELY SAFE PASSCODE THAT IS NEITHER TOO SHORT NOR TOO LONG, WE USED OUR STANDARD BASE OF A SET OF NON CONFUSING CHARACTERS. THAT WAY, WE ENSURE THAT WE CAN MEET THE REQUIREMENT FOR HIGH ACCOUNT NUMBERS, A STAGGERING (6.15×10^{20}) (615 QUINTILLION). THIS ELIMINATED THE NEED FOR CHECKING THE "UNIQUENESS REQUIREMENT" AS THE NUMBER OF COMBINATIONS IS SO LARGE THAT THE COLLISION THRESHOLD IS ENORMOUS.

IN FACT, TO ATTAIN A COLLISION PROBABILITY OF 50%, WE WOULD NEED TO GENERATE 92 BILLION IDS (9 ACCOUNTS FOR EVERY PERSON ON THE PLANET WITH 2 BILLION ACCOUNTS RESERVED FOR THE AUTHOR AND WE STILL WOULDN'T EXHAUST THE TOTAL NUMBER OF EXISTING ID'S)

A QUICK RESEARCH SHOWS THAT EVEN THE LARGEST OF BANKS OPERATE WITH ONLY SINGLE OR DOUBLE DIGIT MILLIONS OF ACCOUNTS WHICH MAKE OUR "MAKESHIFT" ID'S PERFECT FOR THE SITUATION.

-SIMILARLY, WE ADOPTED A DAY/MONTH/YEAR STRUCTURE FORMAT. WE ALSO STANDARDIZED OUR ADDRESSES AND IBANS FORMATS TO TRY AND IMITATE INTERNATIONAL STANDARDS, "TN CN 11 BRC ACCNUM NC".

NOT ONLY THAT, BUT WE ALSO CREATED OUR OWN BRANCH CODE NOTATION, "X00" FOR BRANCH TRACKING PURPOSES. THE EPP CONTAINS NECESSARY INFORMATION ON HOW TO GENERATE AN IBAN, ITS DEPENDENCIES AND THE BRANCH CODE GIVEN INITIAL VALUES. IT ALSO SIGNALS THE MEANING OF EACH OF THEM.

-MOST IMPORTANTLY, WE DECIDED ON THE IMPLEMENTATION OF A USER-FRIENDLY INTERFACE USING QTCREATOR. ARMED WITH THE EPP KNOWLEDGE, WE HAD A CLEAR VIEW OF DIFFERENT WIDGETS THAT WERE AWAITING IMPLEMENTATIONS.
KUDOS FOR LINA FOR WORKING ON THE INTERFACE

IT WAS CLEAR THAT THE PROJECT WAS PROCEEDING SMOOTHLY. WE AGREED ON WHO SHOULD DO WHAT AND PROCEEDED ACCORDINGLY. PREPARATION IS KEY AND ORDER IS A WAY OF LIFE.
THANK YOU FOR CHOOSING SKB.

FINAL TOUCHES()**{**

A LITTLE LESS THAN A WEEK BEFORE THE SUBMISSION DEADLINE, WE HAD MOST OF THE WORK IN PLACE, A FUNCTIONAL INTERFACE AND LOTS OF CODE. AT THIS POINT, WE WERE PUTTING EVERYTHING TOGETHER TO READY FINAL TESTS AND ISSUE THE FINAL DEBUGGING ROUND. AND IN THE SPIRIT OF A TRUE ENGINEER, I ASKED FOR SOME TESTS TO BE DONE BEFORE PUTTING THE WORK TOGETHER. AS EXPECTED, SOME MINOR ISSUES AROSE BUT THEY WERE SWIFTLY AVOIDED.

COUT << **RESULT**

YOU ASKED, AND WE DELIVERED.

WE EVEN HAD TIME TO PRÉPARE THIS NIFTY REPORT. LOOKS COOL, RIGHT?

A MONTH AGO, WE DIDN'T KNOW WHERE TO START. NOW, WE ARE BOASTING ABOUT THE SUCCESSFUL COMPLETION OF OUR PROJECT. AT THE TIME OF WRITING, ALL TASKS WERE COMPLETED, AND THE SOURCE FILES WERE MOSTLY CLEANLY PACKAGED AND READY FOR EVALUATION.

WE PUT A LOT OF EFFORT INTO THIS PROJECT, AND WE TRULY HOPE THAT IT EXCEEDS YOUR EXPECTATION.

KINDLY REFER TO THE SOURCE FILES SENT FOR A FULL HANDS-ON EXPERIENCE OF THE SYSTEM

THANK YOU VERY MUCH

ME AND MY TEAMMATES EMBARKED ON THIS JOURNEY FULL OF CONFIDENCE IN OUR SUCCESS AND WE ARE GLAD THAT IT WAS MORE FRUITFUL THAN WE FIRST IMAGINED. IF ANYTHING, WE ARE THANKFUL FOR PROPERLY MANAGING ROLES.

discussion

AFTER ALL, IT SEEMED THAT THE PROJECT REQUIRED ONLY A CLEAR MINDSET MORE THAN THE ACTUAL CODE. THE FINAL PRODUCT IS AN AMALGAMATION OF CODING, COMMUNICATION, GROUP WORK, AND PERSONAL GROWTH. I PERSONALLY ADMIRER THE SOLIDARITY MY TEAMMATES EXPRESSED TOWARDS ONE ANOTHER. WHEN ALL OF US WERE PUT UNDER THE SAME PRESSURE, WE ALL DID OUR BEST TO PROVIDE SUPPORT AND ADVICE. IT HAS BECOME INHERENTLY CLEARER THAT ITS TRUE PURPOSE WASN'T SOLELY TO TEST OUR KNOWLEDGE, BUT RATHER TO CHALLENGE OUR OWN CONVICTION AND PRINCIPLES. THIS PROJECT SERVED AS OUR FIRST EXPERIENCE AS ENGINEERS IN AN ATTEMPT TO BALANCE WHAT OTHERS DO, AND FINALLY HOW TO CAN MANAGE, WHAT YOU KNOW AND WHAT OTHERS DO, AND FINALLY HOW TO BALANCE YOUR TIME, YOUR TOOLS, AND COMBINE YOUR KNOWLEDGE NOT AS AN INDIVIDUAL, BUT RATHER IN THE SPIRIT OF THE GROUP. CAN EACH AND EVERY ONE OF THE GROUP EXPRESS THEMSELVES AND SHOWCASE THEIR POTENTIAL GIVEN REAL LIFE CONSTRAINTS?

CONCLUSION

TASKED WITH CREATING A FUNCTIONAL BANKING MANAGEMENT SYSTEM,

OUR GROUP GOES ON A MONTH-LONG JOURNEY EXPERIENCING A DAY IN THE LIFE OF AN ENGINEER BY LEARNING HOW TO FACE A PROBLEM, DECONSTRUCT IT AND BUILD UPON ESTABLISHED FACT IN A WELL COMMUNICATED ENVIRONMENT TO OVERCOME SAID CHALLENGE NO MATTER THE PROBLEMS WE FACE ALONG THE WAY. THE COMPLETION OF THE PROJECT REQUIRED ONE TO BE LEVEL-HEADED, SERIOUS, SUPPORTING AND IN DIRECT COMMUNICATION WITH OTHER MEMBERS OF THE GROUP. IN FACT, IT TURNS OUT THAT A LITTLE SUPPORT GOES A LONG WAY.

THANK YOU VERY MUCH

PREPARATION IS KEY AND ORDER IS A WAY OF LIFE.
THANK YOU FOR CHOOSING SKB.

RECOMMENDATIONS:

ONE SHOULD BELIEVE IN HIMSELF AND SHOULD BELIEVE IN HIS TEAMMATES. ON THAT NOTE, CHOOSING THE RIGHT TEAMMATES IS THE FIRST STEP TOWARDS SUCCESS. ONE COULD EVEN SAY THAT CHOOSING THE RIGHT "COMRADES" AMOUNTS IN AND OF ITSELF TO COMPLETING HALF THE PROJECT.

REFERENCES:

MOST RESEARCH WAS DONE REFERENCING (ISO) LAWS AND REGULATIONS FOR STANDARDIZED SYSTEMS. IT HAS TO BE MADE CLEAR THAT THESE REGULATIONS WEREN'T FOLLOWED THROUGH AND THROUGH FOR PRACTICALITY PURPOSES. THIS POINT HAS ONLY BEEN MADE FOR REFERENCE AND TO CLEAR OURSELVES FROM ANY WRONG-CALLINGS. NO LAWS WERE BROKEN IN ANY WAY SHAPE OR FORM. ALL THIS WAS DONE FOR EDUCATIONAL PURPOSES SOLELY.

APPENDICES:

—THIS REPORT SHOULD COME WITH A COPY OF THE "EPP" IN A CORRESPONDING FORMAT.

#PRAGMA EXIT